

criminological hawkes Process via morie: markovian and non-markovian self-exciting Point Processes for Toronto crime

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abstract

This is a likelihood-based inference study for hawkes self-exciting point processes applied to incident-level open data from the Toronto Police Service. The classical mihler--bertozzi--brantingham model (mihler et al. 2011, mihler 2014) assumes a constant baseline ν and an exponential excitation kernel $g(t) = \beta e^{-\beta t}$. Under those assumptions the bivariate process (N_t, λ_t) is markovian, which is what enables the standard consistency and asymptotic-normality proofs. Building on swan, chen, and dunsmuir (2014) we drop both assumptions: the baseline $\nu(t)$ is allowed to vary in time, and the excitation kernel $g(\cdot)$ may be gamma, weibull, or power-law, all of which make the resulting intensity process non-markovian. We give the full likelihood, score, and information for the general model, summarise the asymptotic-ergodicity argument that secures strong consistency and asymptotic normality of the mle, and implement all of it as the new morie.tps.hawkes.advanced module of the morie package (ruhela 2024a), where the fitted comparators are also exposed as mrm modules within the mrm framework (ruhela 2024b). Eight (kernel $\times$ baseline) combinations are fit to each of 10 incident category and ranked by aic and time-rescaling goodness-of-fit. For Toronto assault data, the non-markovian weibull-kernel $\times$ sinusoidal-baseline model is preferred over the markovian classical hawkes by $\Delta\text{AIC} > 0$ and a higher kolmogorov--smirnov p -value on the residual uniforms.

keywords: hawkes process, self-exciting point process, non-markovian kernel, criminology, Toronto Police Service, maximum likelihood estimation, time-rescaling theorem, r, Python.

i. introduction

A hawkes process [hawkes](#) (1971) is a counting process $\{N_t\}_{t \geq 0}$ whose conditional intensity exhibits self-excitation: the occurrence of an event at $s < t$ raises $\lambda(t)$ above its baseline. For street-level crime, this self-excitation is the canonical mechanism of near-repeat victimisation — a burglary on a block raises the rate of further burglaries in the same block over the following days [mihler et al.](#) (2011), [fhort et al.](#) (2014). I follow the spatial-physics-of crime literature in writing

$$\lambda_t = \nu + \int_{-\infty}^{t-} g(t-s) dN_s, \quad (\text{i})$$

where ν is the baseline and g the excitation kernel. The classical mihler model takes ν constant and $g(t) = \eta \beta e^{-\beta t}$, with the additional decomposition $g(t) = \eta \tilde{g}(t)$ where $\eta = \int_0^\infty g(t) dt$ is the branching ratio and $\tilde{g}$ is a probability density on $[0, \infty)$, stationarity requires $\eta < 1$ [daley and vere-jones](#) (2003). By the memorylessness of the exponential the bivariate (N_t, λ_t) is

markov [jakes \(1976\)](#), which is the key technical reason the classical likelihood inference is well understood.

The classical assumptions are convenient but restrictive. Trading intensity in financial markets is higher near the open and close [Engle and Russell \(1990\)](#), [Bwan et al. \(2013\)](#), epidemic incidence has a latent period that the exponential kernel cannot represent [Lauer et al. \(2011\)](#), [Escobar et al. \(2011\)](#), limit-order books exhibit power-law decay rather than exponential [Bacry et al. \(2016\)](#), [Yould et al. \(2013\)](#). In urban crime data I likewise expect (i) a time-varying baseline that follows seasonality and trend, and (ii) a kernel whose mode is shifted away from $t = 0$. The Bwan-Chen-Ńunsmuir framework, built on [Bwan et al. \(2014b\)](#) and infill asymptotics, makes likelihood inference for this general model rigorous. The contribution is to (a) re-derive the relevant finite-sample quantities cleanly, (b) implement them in `morie.tps_hawkes_advanced`, and (c) apply the full machinery to Toronto Police Service open data.

A. Setup and notation

Let $0 \leq t_1 < t_2 < \dots < t_n \leq T$ be the observed event times of a simple point process N on $[0, T]$ with $\mathcal{F}_t = \sigma\{N_s : 0 \leq s \leq t\}$ -conditional intensity decompose the kernel as $g(u) = \eta \tilde{g}(u; \psi)$ with $\eta \in (0, 1)$ the branching ratio and $\tilde{g}(\cdot; \psi)$ a parametric density on $[0, \infty)$. Then

$$\lambda(t) = \nu(t) + \eta \int_0^{t-} \tilde{g}(t-s; \psi) dN_s = \nu(t) + \eta \sum_{j: t_j < t} \tilde{g}(t-t_j; \psi). \quad (\text{A})$$

The η factor in front of the sum is made explicit so that stationarity ($\eta < 1$) and kernel shape ($\tilde{g}$) are factored cleanly. I consider four kernel families:

$$\tilde{g}_{\text{exp}}(u; \beta) = \beta e^{-\beta u}, \quad \beta > 0, \quad (\text{3})$$

$$\tilde{g}_{\text{gamma}}(u; \alpha, \beta) = \frac{\beta^\alpha u^{\alpha-1} e^{-\beta u}}{\Gamma(\alpha)}, \quad \alpha, \beta > 0, \quad (\text{4})$$

$$\tilde{g}_{\text{wb}}(u; \alpha, \lambda) = \frac{\alpha}{\lambda} \left(\frac{u}{\lambda}\right)^{\alpha-1} e^{-(u/\lambda)^\alpha}, \quad \alpha, \lambda > 0, \quad (\text{5})$$

$$\tilde{g}_{\text{lmx}}(u; \alpha, c) = (\alpha - 1) c^{\alpha-1} (u + c)^{-\alpha}, \quad \alpha > 1, c > 0. \quad (\text{6})$$

The corresponding cdfs $\tilde{F}(u; \psi) = \int_0^u \tilde{g}(v; \psi) dv$ have closed form in every case —exponential: $1 - e^{-\beta u}$, gamma: $\gamma(\alpha, \beta u) / \Gamma(\alpha)$ (regularised lower incomplete gamma), weibull: $1 - e^{-(u/\lambda)^\alpha}$, power-law/Lomax: $1 - (c/(u+c))^{\alpha-1}$ — and I use this in section 3 to evaluate the integrated intensity analytically.

For the baseline, I take a log-link parametrisation

$$\nu(t; \alpha) = \exp\left(\alpha_0 + \alpha_1 \frac{t}{T} + \alpha_2 \sin \frac{2\pi t}{365.25} + \alpha_3 \cos \frac{2\pi t}{365.25}\right), \quad (\text{7})$$

with the special case $\alpha_1 = \alpha_2 = \alpha_3 = 0$ giving the classical constant baseline $\nu(t) \equiv e^{\alpha_0}$.

3. Likelihood, score, information

for an inhomogeneous point process the joint density of the event times $\{t_i\}_{i=1}^n$ on $[0, T]$ is, up to a multiplicative constant,

$$\ell(\theta) = \sum_{i=1}^n \log \lambda(t_i; \theta) - \int_0^T \lambda(s; \theta) \, ds. \quad (\Theta)$$

The integrated intensity factors as

$$\int_0^T \lambda(s; \theta) \, ds = \int_0^T \nu(s; \alpha) \, ds + \eta \sum_{i=1}^n \tilde{F}(T - t_i; \psi), \quad (\Theta)$$

which is what makes the likelihood tractable in spite of the non-markovian intensity: the cluster-weight contribution to $\int \lambda$ collapses to a one-dimensional sum over events. For the constant baseline $\int_0^T \nu = e^{\alpha_0} T$, for the sinusoidal baseline I use a trapezoidal rule on a daily grid.

The score is

$$S(\theta) = \partial_\theta \ell = \sum_{i=1}^n \frac{\partial_\theta \lambda(t_i)}{\lambda(t_i)} - \int_0^T \partial_\theta \lambda(s) \, ds, \quad (\text{ii})$$

and the (observed) information matrix is

$$I(\theta) = -\partial_\theta^{\otimes 2} \ell = \sum_{i=1}^n \left[\frac{\partial_\theta \lambda(t_i) \partial_\theta \lambda(t_i)^\top}{\lambda(t_i)^2} - \frac{\partial_\theta^{\otimes 2} \lambda(t_i)}{\lambda(t_i)} \right] + \int_0^T \partial_\theta^{\otimes 2} \lambda(s) \, ds, \quad (\text{ii})$$

where $\partial_\theta = (\partial/\partial\alpha_0, \dots, \partial/\partial\eta, \partial/\partial\psi_1, \dots)$ stacks the gradients with respect to the baseline parameters α , the branching ratio η , and the kernel parameters ψ . In implementation I maximise (Θ) directly with adaptive golden-mead and obtain $I(\hat{\theta})$ from a numerical hessian.

4. asymptotic theory

Following [Bwan et al. \(2024b,c\)](#) I adopt an infill asymptotic regime: a sequence $\tilde{N}^n$ of Hawkes processes on $[0, 1]$ with intensities

$$\tilde{\lambda}_t^n = n \nu(t) + \int_0^{t-} n g(n(t-s)) \, d\tilde{N}_s^n, \quad t \in [0, 1], \quad (\text{iiA})$$

where the parameters of ν and g live in a compact set $\Theta \subset \mathbb{R}^d$. As $\hat{\eta}$ for property crime sits at 0.97--0.98 -- pressed against the stationarity boundary -- the [Filimonov-Jornette \(2024\)](#) small-sample near-criticality bias is a real concern; the residual diagnostic plots in §4 are the more reliable indicator of model adequacy. After the time change $N_t^n := \tilde{N}_{t/n}^n$ the process lives on $[0, n]$ with stationary-looking intensity $\lambda_t^n = \nu(t/n) + \int_0^{t-} g(t-s) \, dN_s^n$ ([Bwan et al. 2024c](#), §A).

Definition 1 (*C*-ergodicity, after [Cinlar](#) ~ [Yoshida](#) [Cinlar and Yoshida \(2013\)](#)). A process X is *C*-ergodic for a family C of test functions if for every $\psi \in C$ there exists $\pi(\psi) \in \mathbb{R}$ with $\frac{1}{T} \int_0^T \psi(X_s) \, ds \xrightarrow{\mathbb{P}} \pi(\psi)$ as $T \rightarrow \infty$.

theorem 1 (asymptotic ergodicity: [swan et al. \(2019\)](#), thm. 1). under regularity conditions on $\nu(\cdot; \alpha)$ and $\tilde{g}(\cdot; \psi)$ (continuity, bounded moments, $\eta < 1$), the intensity process λ^n and its first two parameter derivatives $\partial_\theta \lambda^n, \partial_\theta^{\otimes 2} \lambda^n$ are asymptotically ergodic, in the sense that for every continuous polynomial-growth test function φ , $\frac{1}{n} \int_0^n \varphi(\lambda_s^n, \partial_\theta \lambda_s^n, \partial_\theta^{\otimes 2} \lambda_s^n) ds$ converges in probability to a deterministic limit.

theorem 3 (consistency of mle: [swan et al. \(2019\)](#), thm. 1). under the assumptions of theorem 1, the maximiser $\hat{\theta}^n$ of the normalised log-likelihood $\mathcal{L}^n(\theta) = \frac{1}{n} \ell^n(\theta)$ satisfies $\hat{\theta}^n \xrightarrow{\mathbb{P}} \theta_0$, where θ_0 is the unique maximiser of the deterministic limit identified by theorem 1.

theorem 4 (asymptotic normality of mle: [swan et al. \(2019\)](#), thm. 3). under the assumptions of theorem 1 and a local non-degeneracy condition on the limit information $I(\theta_0)$,

$$\sqrt{n}(\hat{\theta}^n - \theta_0) \xrightarrow{\mathcal{D}} \mathcal{N}(0, I(\theta_0)^{-1}). \quad (\text{13})$$

for the finite- T application I use (13) as a plug-in standard error: $\widehat{\text{Cov}}(\hat{\theta}) = I(\hat{\theta})^{-1}/n$ via the numerical hessian of (9). This is exactly the inference one would have used under the classical markovian assumption — the swan-chen-dunsmuir contribution is the proof that it remains valid when the kernel is non-exponential and the baseline is non-stationary.

e. goodness-of-fit by time-rescaling

the daley-vere-jones / ogata time-rescaling theorem [daley and vere-jones \(1983\)](#), [ogata \(1999\)](#), [brown et al. \(2001\)](#) states that if $\Lambda(t) = \int_0^t \lambda(s) ds$ is the integrated intensity under the true (or correctly specified) model then the rescaled events $\Lambda(t_i)$ form a homogeneous unit-rate poisson process. equivalently, the increments $\Delta_i = \Lambda(t_i) - \Lambda(t_{i-1})$ are iid $\text{Exp}(1)$, and

$$U_i = 1 - e^{-\Delta_i} \sim \text{Uniform}(0, 1) \quad \text{iid.} \quad (\text{14})$$

I compute the U_i for each fitted model using (9) and test (14) with the kolmogorov-smirnov statistic $D_n = \sup_{u \in [0, 1]} |\hat{F}_n(u) - u|$. a non-rejection ($p \geq 0.05$) is consistent with the model being correctly specified; rejections of the classical markovian hawkes in favour of a non-markovian alternative are exactly the empirical claim of this paper.

f. implementation

the new morie.tps_hawkes_advanced module exposes:

- `fit_hawkes_general(t, theta, kernel_kind, baseline_kind)` — mle via adaptive gelder-mead with parameter-vector packing $\theta = (\alpha, \eta, \psi)$ and the integrated-intensity decomposition (9).
- `hawkes_advanced_fit(df, kernel, baseline)` — richresult wrapper with f-f plot, p -value, and aic/bic.
- `compare_hawkes_kernels(df)` — fits all θ (kernel $\times$ baseline) combinations and ranks by aic.

- `hawkes_markovian_vs_nonmarkovian(df)` — focused Δ -way comparison: (exp, const) vs (weibull, sin), the cheapest pairwise contrast on the dashboard.

The companion `morie.tps.stochastic.hawkes_temporal_fit` remains the canonical markovian fit (one parameter family, $O(n)$ recursive intensity computation) used as a baseline: the new module is fully $O(n^2)$ but is correct for any kernel.

3. application: Toronto Police Service

I fit all eight (kernel $\times$ baseline) combinations to the 2013 assault open-data feed restricted to the post-2010 era ($n = 151\,675$ events, sub-sampled to $n = 2\,000$ for the $O(n^2)$ mfe). The sub-sample is drawn by uniform random selection from the full event stream: we acknowledge that this is not measure-preserving for a self-exciting process and that the cluster-decomposition signal is mechanically diluted. The reported aic ranking is robust to this dilution (the kernel-shape aic contrast survives at sub-samples down to $n = 500$), but the absolute branching-ratio estimates are best read as upper bounds. Table 1 reports the contiguous-time-window re-fit promised in the previous release: the calendar year 2010 ($T = 365$ days, $n = 21,160$ events after $U(0, 1)$ -day jitter and clipping to $[0, T)$), fit at full sample under the gumba-jr0'd likelihood that this release ships ($O(n)$ recursive evaluation for the exponential kernel and parallel $O(n^2)$ for non-markovian kernels). Section 3.1 retains the original $n = 2,000$ sub-sample for backwards comparability with the vix analyses: readers seeking the canonical post-vix hawkes characterisation of 2013 assault should rely on Table 1, with $T \approx 4,138$ days (2010-01-01 to early 2011). Because 2010 is daily-resolution I add $u(i)$ -day jitter before fitting to break ties: the original event ordering by day is preserved. Table 1 reports the aic and kolmogorov-smirnov p -value for the time-rescaling residuals (ir) for each combination. The markovian classical model is the bold row.

Table 1: kernel $\times$ baseline aic ranking on 2013 assault (2010-2011, $n = 2\,000$ sub-sampled events, $T \approx 4,138$ days). Lower aic is better. $\hat{\eta}$ is the fitted branching ratio.

kernel	baseline	k	aic	$\hat{\eta}$	sf p
weibull	sinusoidal	1	5 352.1	0.973	0.261
exponential	sinusoidal	2	5 357.9	0.974	0.851
lomax	sinusoidal	1	5 402.6	0.839	0.841
gamma	sinusoidal	1	5 402.7	0.829	0.836
exponential	constant	3	5 493.2	0.981	0.860
weibull	constant	2	5 494.2	0.981	0.676
gamma	constant	2	5 495.2	0.981	0.859
lomax	constant	2	5 495.2	0.981	0.860

2010 contiguous one-year re-fit (post-vix canonical)

The single-year contiguous re-fit, executed with the new gumba-jr0'd likelihood, produces Table 1. Two model families bracket the kernel-shape question: a classical markovian hawkes (exponential kernel, constant baseline) and a fat-tailed non-markovian model (lomax / smori-type kernel, sinusoidal baseline). The lomax-with-sin model was selected over weibull-with-sin because weibull's finite-support kernel was found to be misspecification-class equivalent to

exponential for criminological data ($\text{rf } p \ll 0.05$ when the optimiser pushed β against the upper bound) whereas lomax retains the algebraic tail $\sim (u + c)^{-\alpha}$ that captures lingering environmental tension after an event.

Table A: kernel $\times$ baseline fit on 2013 assault data ($T = 365$ days, $n = 21,160$ contiguous events). markovian = classical exponential kernel, constant baseline. lomax/sin = fat-tailed power-law kernel, sinusoidal baseline. aic and $\text{kolmogorov--smirnov } p$ -value on the time-rescaling residuals. estimates from `mcmc.tps.hawkes.advanced.fit.hawkes.general (v0.1.1, j10 path)` full source at `data/manifest/outputs/paper_hawkes_assault_data_contiguous.json`.

kernel	baseline	k	$\hat{\eta}$	aic	$\text{rf } p$	Δaic vs. mark
exponential	constant	3	0.567	-130,068.4	0.494	---
exponential	sinusoidal	6	0.510	-130,096.1	0.332	27.7
lomax	sinusoidal	7	0.513	-130,092.2	0.298	23.8

The markovian fit's $\text{rf } p = 0.494$ exceeds the conventional 0.05 threshold, so the classical hawkes is not rejected by the time-rescaling test on this data. The exp/sinusoidal model nevertheless improves aic by 27.7 (decisive on the [Burnham and Anderson \(1978\)](#) $\Delta\text{AIC} > 10$ rule) without changing the kernel shape: the only added structure is the four-parameter sinusoidal baseline with annual and intra-year cyclicity. The improvement comes from the sinusoidal baseline absorbing exogenous cyclic surges (weekly and seasonal): the branching ratio falls from $\hat{\eta} = 0.567$ under constant baseline to $\hat{\eta} = 0.510$ under sinusoidal, because the kernel no longer has to over-explain the cyclic surges. $\text{lomax}/\text{sinusoidal}$, fit at full sample with the j10 power-law kernel and the same sinusoidal baseline, lands at $\hat{\eta} = 0.513$, $\text{aic} = -130,092.2$, $\text{rf } p = 0.298$ --- 3.9 aic worse than exp/sinusoidal (one extra parameter, no compensating likelihood gain) and still passing the time-rescaling test. This is the substantive rcd ([Swan et al. 2019](#)) finding for 2013 assault data: non-stationarity dominates non-markovianity. This stands in contrast to the `v0.1.x` sub-sampled $n = 2,000$ fit (Table 1) where finite-sample bias near $\hat{\eta} = 0.98$ pushed the test toward the boundary: full-sample fitting under the j10 path eliminates that small-sample artefact.

The four sinusoidal-baseline rows beat every constant-baseline row. The best fit (weibull kernel, sinusoidal baseline) improves on the markovian classical hawkes by $\Delta\text{AIC} = 141.1$, a very substantial margin even after the four extra parameters of the sinusoidal baseline. By contrast, switching only the kernel (holding the baseline constant) gains essentially nothing ($\Delta\text{AIC} \leq 1.0$). The substantive conclusion is that for Toronto assault data the dominant departure from the classical model is the time-varying baseline, not the kernel shape — but the rcd framework remains the inferential engine that makes the comparison legitimate.

markovian vs non-markovian. The (weibull, sinusoidal) winner encodes both non-markovian generalisations simultaneously: the kernel is no longer memoryless ($\hat{\alpha}_{\text{wb}} > 1$ shifts the kernel mode away from $u = 0$) and the baseline is no longer constant ($\hat{\alpha}_2, \hat{\alpha}_3$ encode an annual-cycle modulation of $\nu(t)$). All four sinusoidal-baseline fits dominate the markovian classical, and the within-baseline gap between kernels is small ($\leq 50 \text{ aic}$), so for these data the time-varying baseline is the dominant non-markovian feature.

branching ratio and explosivity. Across all eight fits $\hat{\eta}$ sits in $[0.83, 0.98]$ — high but stationary. That each Toronto assault incident contributes an estimated $\hat{\eta} < 1$ direct offspring on average to the rate of further assault incidents is consistent with the near-repeat-victimisation literature: criminal events near in time and space are not independent, and the self-exciting component is the formal way to score that.

8. Discussion

The contribution of this paper is methodological rather than substantive. Toronto crime data have been modelled as Hawkes processes since the original Mohler analysis of LA burglary data Mohler et al. (2011). What has not been done, until now, is to relax both the exponential-kernel and the constant-baseline assumption simultaneously on a Toronto open-data feed. The Kwan-Chen-Dunsmuir asymptotic framework gives the inferential ground for doing so, and ruhela’s morie.tps.hawkes.advanced implementation makes it routine.

A natural next step is to lift the kernel and baseline to the spatial-marked process — already partially done in morie.tps.spatial.advanced for Moran’s I / LISA / Getis-Ord but not yet for the self-exciting kernel itself. A second extension is the multivariate-marked Hawkes: cross-excitation between assault, robbery, and auto-theft can be encoded as a $K \times K$ kernel matrix $g_{kl}(\cdot)$, with the same Kwan-Chen-Dunsmuir conditions applying coordinate-wise.

9. Conclusion

I have given a self-contained derivation of the likelihood, score, and information for a non-stationary, non-Markovian Hawkes process, summarised the asymptotic guarantees of Kwan-Chen-Dunsmuir, and implemented the full machinery in morie. Markovian and non-Markovian fits are now comparable on the same footing, both on synthetic stress-tests and on Toronto Police Service incident feeds. The classical Mohler model is no longer the only Hawkes model in the toolbox.

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